

SANTA ROSA DE COPAN, Honduras

WMO: 787170

Lat: 14.783N

Lon: 88.783W

Elev: 3540

StdP: 12.91

Time Zone: -6.00 (W06)

Period: 83-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	51.2	53.3	47.2	54.7	64.1	49.7	60.2	64.0	16.4	70.0	14.5	69.4	1.2	0	0.264

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	21.2	87.9	67.6	86.0	67.2	84.1	67.4	72.4	78.9	71.6	78.2	70.9	77.6	6.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.6	128.8	74.2	69.9	125.7	73.5	69.2	122.7	72.9	38.6	79.2	37.8	77.9	37.1	77.8	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.9	12.6	10.9		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	69.8	64.8	67.4	69.1	73.0	72.9	72.7	71.7	72.1	72.0	70.0	66.6	64.7
	DBStd	4.56	4.26	4.50	5.06	3.74	3.08	1.69	1.86	1.44	1.75	3.10	4.12	3.90
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	201	57	25	24	4	0	0	0	0	0	5	33	53
	CDD50	7209	457	488	592	691	709	680	673	684	661	618	499	457
	CDD65	1934	49	93	150	245	244	230	208	219	211	158	82	45
	CDH74	9588	205	624	1170	1725	1574	1048	789	841	784	480	204	144
	CDH80	1788	21	131	341	539	410	124	55	47	71	29	10	10
Wind	WSAvg	3.8	3.8	4.2	4.6	4.6	3.9	3.4	4.1	3.5	3.3	3.4	3.4	3.4
Precipitation	PrecAvg	64.4	1.6	1.1	1.1	2.0	5.5	10.8	7.1	9.2	10.3	6.7	3.9	2.1
	PrecMax	78.8	2.6	3.3	5.1	6.6	9.3	21.4	18.9	16.5	18.3	15.3	8.7	4.7
	PrecMin	51.2	0.9	0.2	0.1	0.2	2.5	4.6	1.9	3.3	4.1	1.9	2.2	0.9
	PrecStd	7.5	0.4	0.6	1.0	1.7	1.9	3.6	3.5	3.5	3.6	2.8	1.7	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.4	88.0	89.6	89.8	89.9	86.1	83.8	82.7	84.3	82.6	81.0	81.3
		MCWB	64.8	67.2	65.8	68.3	66.9	69.5	67.8	69.3	68.9	70.3	68.7	67.6
	2%	DB	78.9	83.8	86.4	87.9	86.8	82.8	81.0	81.0	82.2	80.7	78.7	77.4
		MCWB	64.6	65.7	65.8	68.0	67.8	69.3	67.6	68.6	69.5	69.3	67.7	65.8
	5%	DB	76.3	80.6	84.1	85.9	84.4	81.6	79.9	80.2	80.6	78.9	76.7	75.3
		MCWB	64.1	64.6	65.7	67.4	67.8	69.1	67.8	68.5	69.3	68.9	67.0	65.4
	10%	DB	73.6	77.4	80.8	83.8	82.2	80.3	78.6	78.8	78.9	77.0	74.4	73.1
		MCWB	63.9	64.3	65.5	67.5	67.7	68.9	68.1	68.5	69.0	68.4	66.6	64.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.2	70.2	70.9	72.9	73.1	73.7	72.6	72.2	72.9	72.3	71.2	70.2
		MCDB	75.3	81.9	78.8	79.2	78.2	79.6	78.3	78.8	79.0	79.5	78.3	77.6
	2%	WB	66.5	68.1	69.3	71.1	71.4	72.0	71.1	71.0	71.5	70.9	69.2	67.6
		MCDB	73.1	77.1	76.9	78.5	77.7	78.6	76.9	77.5	77.8	77.9	75.0	73.9
	5%	WB	65.7	66.9	68.2	70.3	70.5	71.0	70.2	70.3	70.7	70.0	68.3	66.5
		MCDB	71.9	75.0	76.1	78.0	77.3	77.6	76.0	76.7	77.0	76.3	73.6	72.4
	10%	WB	64.8	65.9	67.2	69.3	69.5	70.3	69.3	69.5	69.8	69.1	67.4	65.5
		MCDB	70.6	73.3	75.4	77.3	76.7	76.6	74.9	75.6	76.2	74.7	72.2	70.7
Mean Daily Temperature Range	5% DB	MDBR	14.7	18.9	20.5	21.2	19.8	16.4	15.2	15.3	15.3	13.2	12.7	13.4
		MCDBR	21.1	25.4	26.2	24.6	22.9	18.5	16.7	16.6	17.3	16.2	17.2	18.9
		MCWBR	8.7	9.8	9.1	8.0	7.6	7.0	6.5	6.4	7.0	7.1	7.8	8.5
	5% WB	MCDBR	17.5	21.8	22.6	22.0	19.8	16.4	15.4	15.5	15.9	14.9	14.5	16.1
		MCWBR	7.9	9.2	8.9	8.1	7.3	7.2	7.0	6.8	7.4	7.0	7.2	8.0
Clear-Sky Solar Irradiance	taub		0.325	0.332	0.358	0.412	0.445	0.392	0.394	0.383	0.378	0.366	0.342	0.332
	taud		2.602	2.586	2.484	2.315	2.259	2.466	2.454	2.472	2.484	2.543	2.588	2.595
	Ebn at Noon		309	311	303	284	271	283	283	289	292	295	301	304
	Edh at Noon		30	31	36	43	44	36	36	36	35	33	30	29
All-Sky Solar Radiation	RadAvg		1395	1669	1847	1859	1685	1690	1804	1816	1657	1421	1352	1339
	RadStd		100	102	110	106	116	96	102	97	101	141	81	88

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page